ANALYTICAL CUMULATIVE EXAM

April 25, 1992

There are three natural decay chains.

One beginning with ^{238}U and ending with ^{206}Pb.

One beginning with ^{232}Th and ending with ^{208}Pb.

One beginning with ^{235}U and ending with ^{207}Pb.

These chains have significance for many analytical, geochemical, and environmental studies.

Write an essay on ONE of these studies.

Credit will be given for accuracy, breadth, and clarity.

Cu 63.546 σ_a 3.77, 4.1			Cu58 1+ 3.21s β^+ 7.4,… γ 1.454, 1.448,… E 8.563	Cu59 3/- 81.8s .75,…, < .014, .878,… .01	Cu60 2+ 23.0m β^+ 2.00,… γ 1.3325, 1.792,… E 6.127	Cu61 3/- 3.37h β^+ 1.22,…, < γ .283, .656,… E 2.238	Cu62 1+ 9.74m β^+ 2.93,…, < γ 1.1730, .8757,… E 3.95	Cu63 3/- 69.2 σ_γ 4.4, 4.9 62.929599	Cu64 1+ 12.71h <, β^- .578 β^+ .653 γ 1.346 E$^-$ .578 E$^+$ 1.675	Cu65 3/- 30.8 σ_γ 2.17, 2.2 64.927792	Cu66 1+ 5.10m β^- 2.63,… γ 1.039,… σ_γ 1.3, 1.02, 5.3, 10 E 2.642	Cu67 3/- 61.7h β^- .39,… γ .1846, .09330,… E .58	(6-) Cu68 1+ 3.8m IT .111, .637 γ .526, .084,… β^- 1.7, 1.8	31s β^- 3.5,… γ 1.078,… E ~4.62	Cu69 (3/-) 3.0m β^- 2.5,… γ 1.007, .834,… E 2.5	(7-) Cu70 (1+) 48s β^- 3.3, 4.5,… γ .885, .902, 1.252,…	5s β^- 5.3, 6.1 γ .885,… E 6.2	
Ni54 33.9579	Ni55 <5s 54.95133	Ni56 6.10d <, γ .1583, .8118,… E 2.14	Ni57 3/- 36.1h <, β^+ .84,… γ 1.378,… E 3.24	Ni58 68.3 .6, 2.2 7mb 57.935347	Ni59 3/- 8 x 10^{4}a < no γ σ_γ .78, 1.2 x 10^4 σ_α .14, .21 σ_p 2, 3 E 1.073	Ni60 26.1 σ_γ 2.8, 1.5 59.930789	Ni61 3/- 1.1 σ_γ 2.4, 2.3 σ_α < 2 mb 60.931059	Ni62 3.6 σ_γ 14.6, 6.7 61.928346	Ni63 1/- 100a β^- .0659 no γ σ_γ .23 E .0659	Ni64 0.9 σ_γ 1.49, 1.0 63.927968	Ni65 5/- 2.520h β^- 2.13,… γ 1.4818, 1.1155,… σ_γ .24, 1.1 E 2.137	Ni66 54.6h β^- .20 no γ E .24	Ni67 17s β^- 3.8,… γ 1.072, 1.654, .100-1.975 E 3.8	Ni68 67.93187		
Co53 0.25s β^+ p1.56	0.26s β^+	(7)+ Co54 0+ 1.46m β^+ 4.3 γ 1.130, 1.407, .411	0.193s β^+ 7.22 E 8.242	Co55 7/- 17.5h β^+ 1.51,…, < γ .931,… E 3.455	Co56 4+ 78.5d <, β^+ 1.46,… γ .84675, 1.2383, … E 4.568	Co57 7/- 271d .06, .01441, .47,…	5+ Co58 2+ 9.1h IT .0249 σ_γ 1.4 x 10^4	70.8d <, β^+ .475, γ .81076, σ_γ 1.9 x 10^3 7 x 10^3 E 2.308	Co59 7/- 100 σ_γ (20 + 17), 75 58.933198	2+ Co60 10.48m IT .0586, β^- 1.55, γ 1.3326, σ_γ 60, 2 x 10^3	5.27a β^- .318, γ 1.3325, 1.1732,… σ_γ 2.0, 4. E 2.8236	Co61 7/- 1.650h β^- 1.22,… γ .0674,… E 1.322	(5)+ Co62 (2)+ 13.9m β^- 2.9,… γ 1.173, 1.164,…	1.51m β^- 4.1,… γ 1.173,… E 5.24	Co63 (7/-) 27s β^- 3.6,… γ .0871,… E 3.66	Co64 1+ 0.4s β^- ~ 7.0 E 7.31					42
Fe52 0.9m β^+ γ	8.28h β^+ .80, < γ .169, .3780 E 2.37	Fe53 (7/-) 2.53m γ .7010, 1.328, 1.011,…	8.53m β^+ 2.8, 2.4,…, < γ .3779, … E 3.743	Fe54 5.8 δ_γ 2.3, 1.3 53.939612	Fe55 3/- 2.7a < no γ E .231	Fe56 91.8 .6, 1.4 55.934939	Fe57 1/- 2.1 δ_γ 2.5, 1.5 56.935396	Fe58 0.3 σ_γ 1.16, 1.2 57.933278	Fe59 3/- 44.6d β^- .467, .273,… γ 1.0992, 1.2916,… E 1.566	Fe60 ~10^{5}a β^- .134 γ .05860, .0277 E .21	Fe61 (3/-) 6.0m β^- 2.7, 2.6,… γ 1.205, 1.027,… E 3.9	Fe62 68s β^- 2.5 γ .506 E 2.6					40	
Mn51 5/- 46.2m β^+ 2.2,…, < γ .7491,… E 3.208	2+ Mn52 6+ 21.4m β^+ 2.63, < γ 1.4342, … IT .378 E 5.09	5.63d β^+, < β^+ .574 γ 1.4342, .9356, .7442,… E 4.711	Mn53 7/- 3.8x10^{6}a < no γ δ_γ 1.7 x 10^1, 3 x 10^1 E .596	Mn54 3+ 312.5d < γ .8348 δ_γ <10 E 1.377	Mn55 5/- 100 .3, 14.2 54.938046	Mn56 3+ 2.580h β^- 2.85,… γ .84675,… E 3.695	Mn57 (5/-) 1.59m β^- 2.55,… γ .12206, .01441,… E 2.69	(1+) Mn58 (3+) ? 3.0s β^- 6.1	65s β^- 3.8,… γ .81076, 1.3231,… E 5.9	Mn59 (5/-) 4.6s β^- 4.5,… γ .726, .473, .571,… E 5.2							38	
Cr50 4.35 σ_γ 15.8, 7.8 49.946046	Cr51 7/- 27.71d < γ .32008 E .751	Cr52 83.79 δ_γ .8, .5 51.940510	Cr53 3/- 9.50 δ_γ 18., 9. 52.940651	Cr54 2.36 .5, .18 53.938882	Cr55 3/- 3.55m β^- 2.50,… γ 1.528, E 2.604	Cr56 5.9m β^- 1.5 γ .026, .083 E 1.64	Cr57 21s β^- γ .083, .206, .850 E ~4.7 56.9433										36	17.7
V49 331d < no γ E .602	V50 6+ 0.25 > 4x10^{16}a γ .1, .14 σ_γ .41, .64 β^- 1.1 E$^+$ 2.213 γ 1.2 E-1.039 49.947161	V51 7/- 99.75 σ_γ 4.91, 2.7 50.943963	V52 3+ 3.76m β^- 2.5,… γ 1.4342,… E 3.977	V53 7/- .55m β^- .5,… γ .53,… E	V54 (3+) 43s β^- 3.0,… γ .835, .986, 2.25,… E 7.0	V55 ~6.5s β^- γ E ~6.1													
Ti48 73.7 σ_γ 7.9, 3.7 47.947947	Ti49 7/- 5.4 σ_γ 2.2, 1.3 48.947871	Ti50 5.2 σ_γ .177, .11 49.944786	Ti51 3/- 5.75m β^- 2.14,… γ .3201,… E 2.466	Ti52 .7m β^- γ .017 E	Ti53 (3/-) 33s β^- 3.1,… γ .1276, .2284, 1.676, .1008,… E 5.0													34	
Sc47 7/- 3.41d β^- .441, .600 γ .1594 E .601	Sc48 6+ 43.7h β^- .65,… γ 1.3121, .9835, 1.0375,… E 3.99	Sc49 (7/-) 57.3m β^- 2.00,… γ 1.762, 1.623 E 2.004	Sc50 (5)+ 1.71m β^- 3.69,… γ 1.554, 1.121, .524,… E 6.89	Sc51 (7/-) 12.4s β^- 4.4 γ .7, 2.144,… E															32
Ca46 0.0035 δ_γ .7, .3 45.953689	Ca47 (7/-) 4.54d β^- .69,… γ 1.297,… E 1.988	Ca48 0.187 δ_γ 1.1, .5 47.952532	Ca49 (3/-) 8.72m β^- 2.1,… γ 3.084,… E 5.268	Ca50 14s β^- 2 γ 1.591 E															
K45 3/+ 17m β^- 2.1,… γ .174, 1.706,… E 4.20	K46 (2-) 1.8m β^- 6.3,… γ 1.347, 3.70,… E 7.72	K47 1/+ 18s β^- 4.05,… γ 2.013, .586,… E 6.65	K48 (2-) 6.8s β^- 5.0,… γ 3.832, .780,… E12	K49	K50 ~0.4s														
Ar44 11.9m β^- γ 1.705, .182, 1.887,… E3.54 43.96336	Ar45 44.9681	Ar46 45.9681																	30
Cl43																			